
OBJECT ORIENTED PROGRAMMING

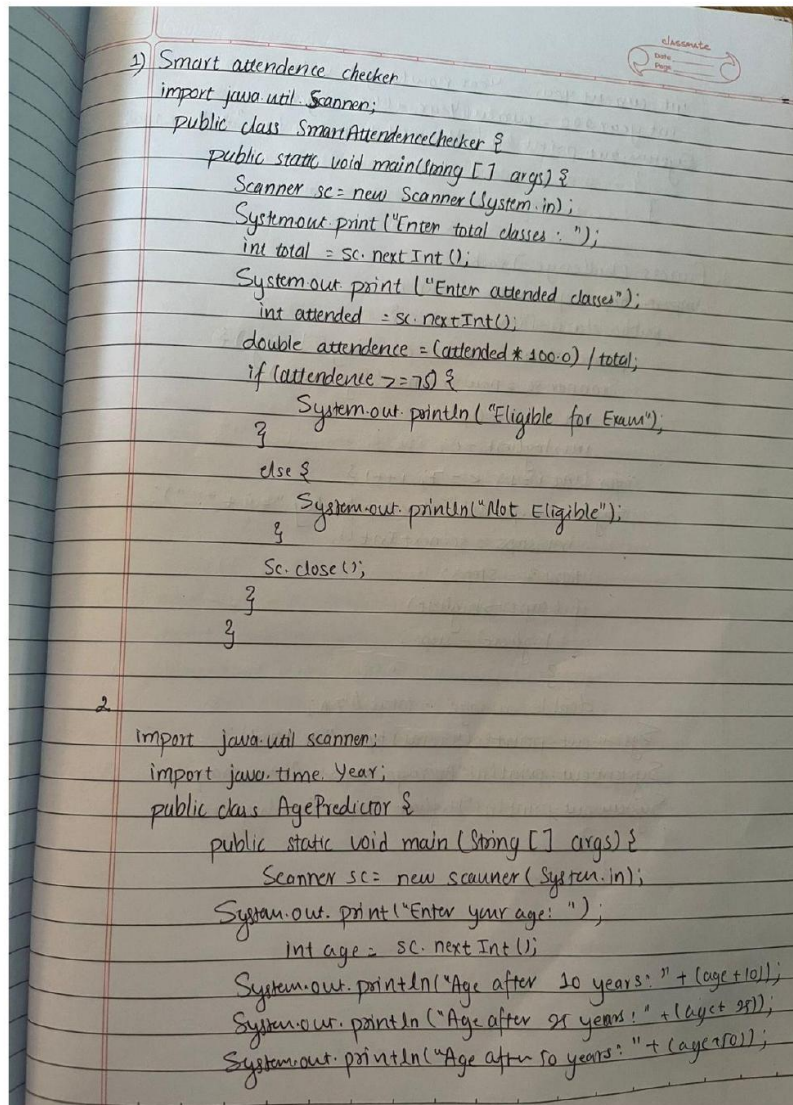
OOP TUTORIAL 2

Java Programming Exercises | Handwritten Code & Program Output

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PART A

Handwritten Java Source Code



The image shows a handwritten Java source code on lined paper. The first program, titled '1) Smart attendance checker', imports java.util.Scanner and defines a class SmartAttendanceChecker with a main method. It prompts the user to enter total classes and attended classes, calculates the attendance percentage, and prints 'Eligible for Exam' if the attendance is 75% or higher, otherwise 'Not Eligible'. The second program, titled '2', imports java.util.Scanner and java.time.Year, and defines a class AgePredictor with a main method. It prompts the user to enter their age and prints the age after 10, 25, and 50 years.

```
1) Smart attendance checker
import java.util.Scanner;
public class SmartAttendanceChecker {
    public static void main(String [] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter total classes : ");
        int total = sc.nextInt();
        System.out.print("Enter attended classes");
        int attended = sc.nextInt();
        double attendance = (attended * 100.0) / total;
        if (attendance >= 75) {
            System.out.println("Eligible for Exam");
        }
        else {
            System.out.println("Not Eligible");
        }
        sc.close();
    }
}

2
import java.util.Scanner;
import java.time.Year;
public class AgePredictor {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print("Enter your age: ");
        int age = sc.nextInt();
        System.out.println("Age after 10 years: " + (age + 10));
        System.out.println("Age after 25 years: " + (age + 25));
        System.out.println("Age after 50 years: " + (age + 50));
    }
}
```

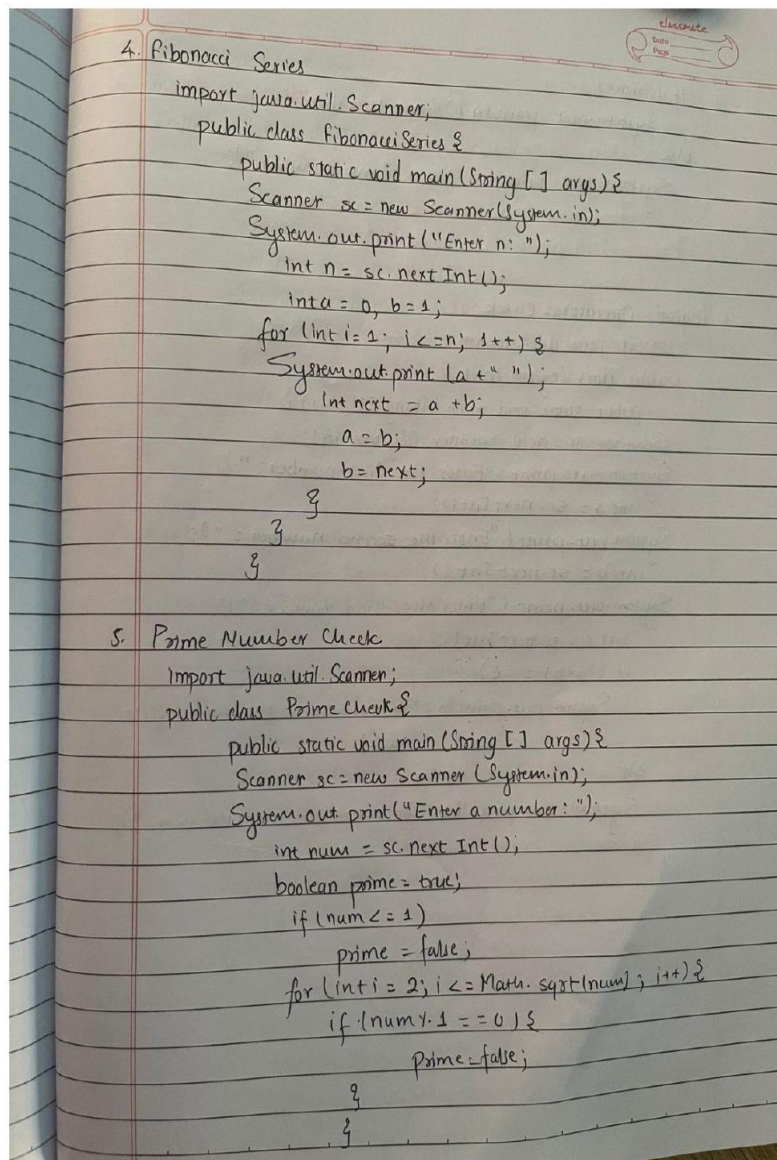
Code Sheet 1 of 9

```
int currentYear = Year.now().getValue();
int year100 = currentYear + (100 - age);
System.out.println("You will turn 100 in: " + year100);
}
}
```

3. Fitness Challenge Tracker

```
import java.util.Scanner;
public class FitnessTracker {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int total = 0;
        int highest = 0;
        for (int i = 1; i <= 7; i++) {
            System.out.println("Enter steps for day " + i + ": ");
            int steps = sc.nextInt();
            total += steps;
            if (steps > highest)
                highest = steps;
        }
        double average = total / 7.0;
        System.out.println("Total steps: " + total);
        System.out.println("Average steps: " + average);
        System.out.println("Highest steps: " + highest);
    }
}
```

Code Sheet 2 of 9



4. Fibonacci Series

```
import java.util.Scanner;
public class FibonacciSeries {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter n: ");
        int n = sc.nextInt();
        int a = 0, b = 1;
        for (int i = 1; i <= n; i++) {
            System.out.print(a + " ");
            int next = a + b;
            a = b;
            b = next;
        }
    }
}
```

5. Prime Number Check

```
import java.util.Scanner;
public class PrimeCheck {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int num = sc.nextInt();
        boolean prime = true;
        if (num <= 1)
            prime = false;
        for (int i = 2; i <= Math.sqrt(num); i++) {
            if (num % i == 0) {
                prime = false;
            }
        }
    }
}
```

Code Sheet 3 of 9

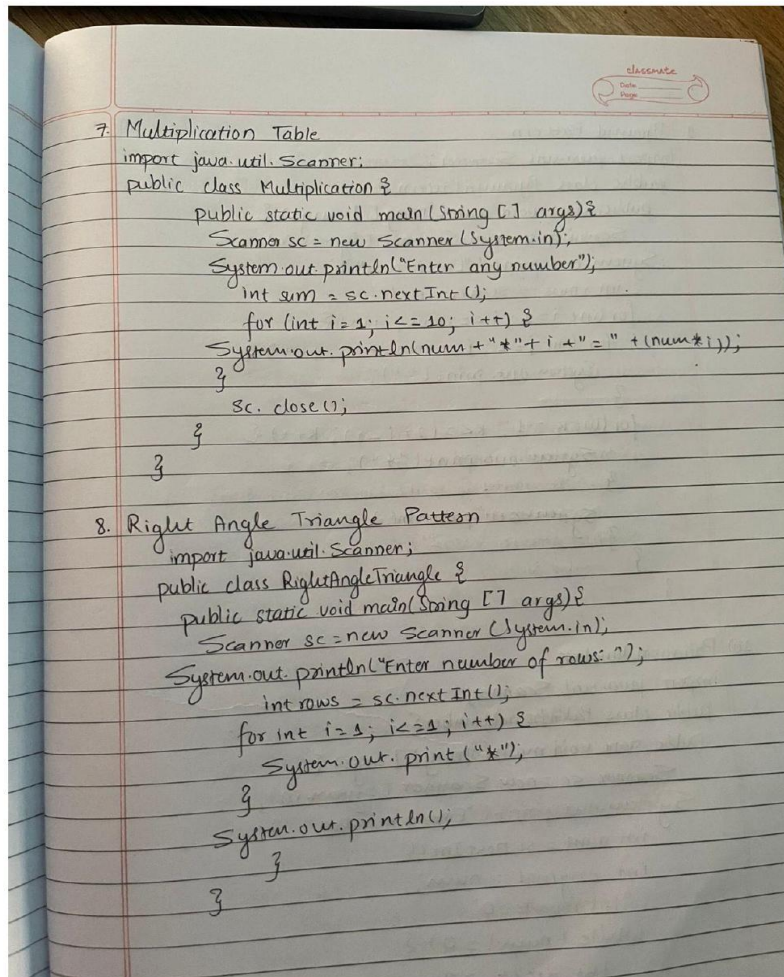
classmate
Date _____
Page _____

```
if (Prime)
    System.out.println("Prime Number");
else
    System.out.println("Not a prime Number");
}
```

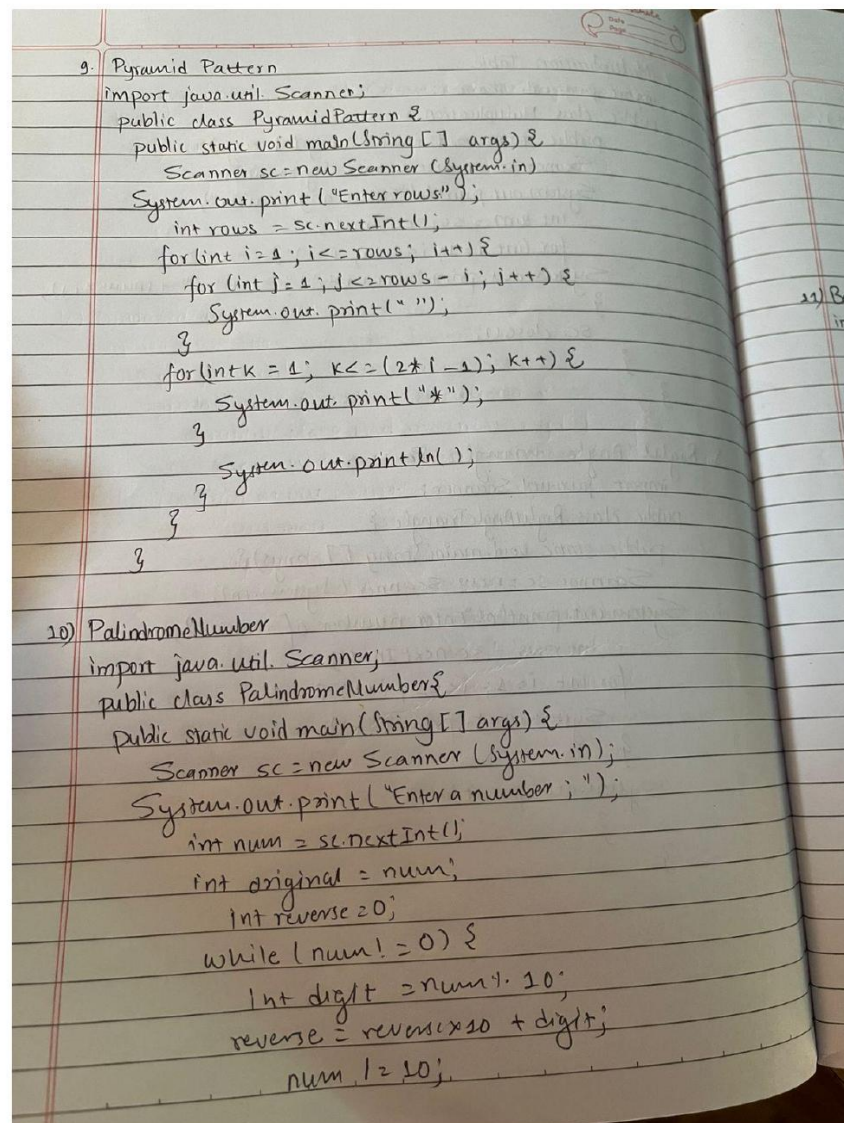
6. Logical Operators Check

```
import java.util.Scanner;
public class LogicalCheck {
    public static void main (String [] args) {
        Scanner sc = new Scanner (System.in);
        System.out.print("Enter the first number: ");
        int a = sc.nextInt();
        System.out.print("Enter the second number: ");
        int b = sc.nextInt();
        System.out.print("Enter the third number: ");
        int c = sc.nextInt();
        if ((a+b) == c)
            System.out.println("Third number is the sum of first two.");
        else
            System.out.println("Third number is not the sum.");
    }
}
```

Code Sheet 4 of 9



Code Sheet 5 of 9



Code Sheet 6 of 9

```

    }
    if (original == reverse)
        System.out.println("Palindrome");
    else
        System.out.println("Not Palindrome");
    }
}

11) Basic Calculator
import java.util.Scanner;
public class BasicCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number :");
        double a = sc.nextDouble();
        System.out.print("Enter second number :");
        double b = sc.nextDouble();
        System.out.print("Enter operator (+, -, *, /) :");
        char op = sc.next().charAt(0);
        switch (op) {
            case '+':
                System.out.println("Result = " + (a+b));
                break;
            case '-':
                System.out.println("Result = " + (a-b));
                break;
            case '*':
                System.out.println("Result = " + (a*b));
                break;
            case '/':
                if (b != 0)
                    System.out.println("Result = " + (a/b));
        }
    }
}
```

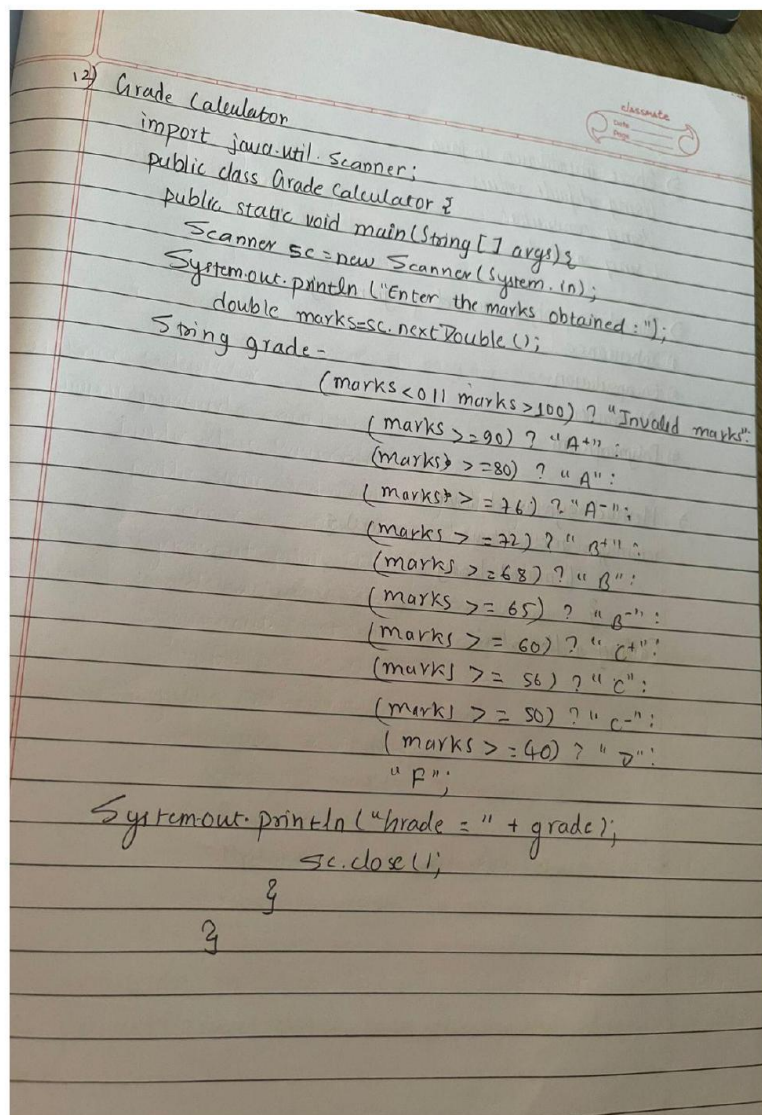
Code Sheet 7 of 9

```
else
    System.out.println("Cannot divide by zero");
break;
default:
    System.out.println("Invalid operator");
}
}
}
```

12) Grade Calculator

```
import java.util.Scanner;
public class Grade-Calculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("Enter marks:");
        int marks = sc.nextInt();
        String grade =
            (marks >= 90) ? "A+" :
            (marks >= 80) ? "A" :
            (marks >= 70) ? "B+" :
            (marks >= 60) ? "B" :
            (marks >= 50) ? "C+" :
            (marks >= 40) ? "C" :
            (marks < 40) ? "D" :
            (marks < 30) ? "E" :
            (marks < 20) ? "F" :
            (marks < 10) ? "G" :
            (marks < 0) ? "Invalid" :
            null;
```

Code Sheet 8 of 9



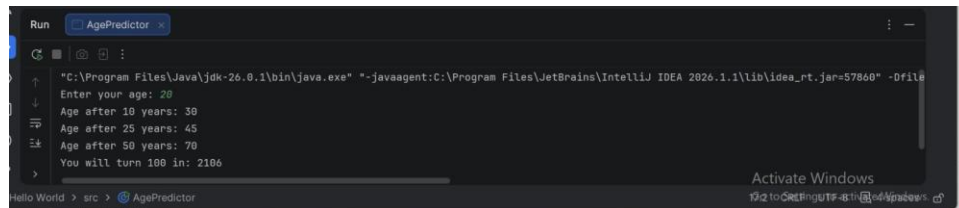
```
12) Grade Calculator
import java.util.Scanner;
public class Grade Calculator {
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter the marks obtained:");
        double marks=sc.nextDouble();
        String grade="";
        (marks<0 || marks>100) ? "Invalid marks" :
        (marks >= 90) ? "A+" :
        (marks >= 80) ? "A" :
        (marks >= 76) ? "A-" :
        (marks >= 72) ? "B+" :
        (marks >= 68) ? "B" :
        (marks >= 65) ? "B-" :
        (marks >= 60) ? "C+" :
        (marks >= 56) ? "C" :
        (marks >= 50) ? "C-" :
        (marks >= 40) ? "D" :
        "F";
        System.out.println("Grade = " + grade);
        sc.close();
    }
}
```

Code Sheet 9 of 9

PART B

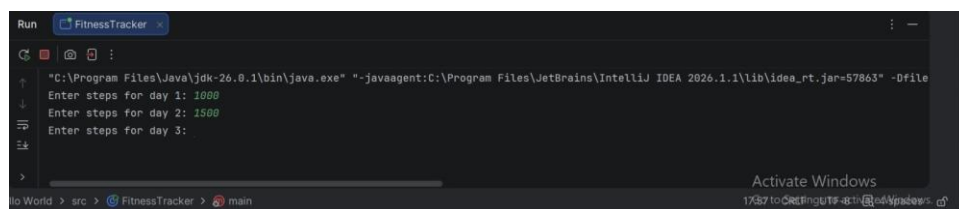
Program Outputs (IntelliJ IDEA Console)

1. Age Predictor



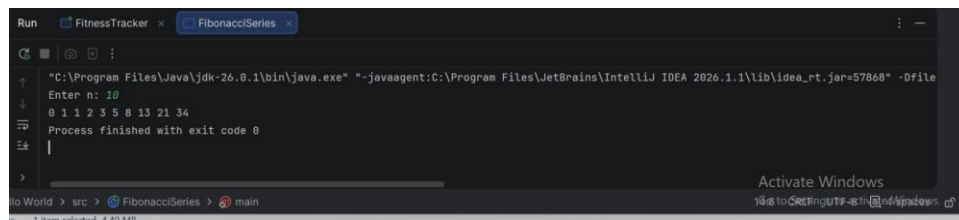
```
Run AgePredictor
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57860" -Dfile.encoding=UTF-8
Enter your age: 20
Age after 10 years: 30
Age after 25 years: 45
Age after 50 years: 70
You will turn 100 in: 2106
Process finished with exit code 0
```

2. Fitness Challenge Tracker (sample run)



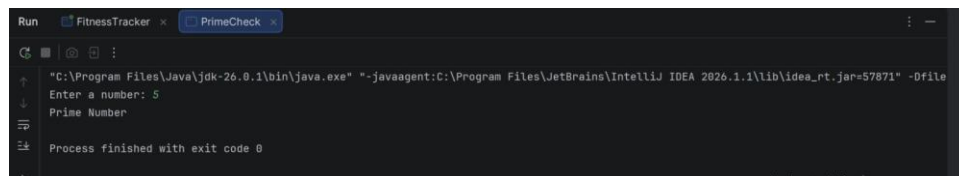
```
Run FitnessTracker
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57863" -Dfile.encoding=UTF-8
Enter steps for day 1: 1000
Enter steps for day 2: 1500
Enter steps for day 3: 1000
Process finished with exit code 0
```

3. Fibonacci Series



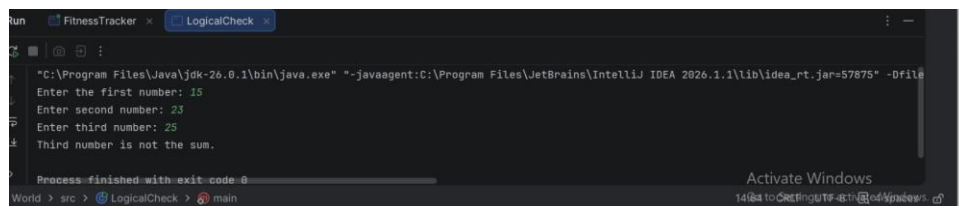
```
Run FitnessTracker FibonacciSeries
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57868" -Dfile.encoding=UTF-8
Enter n: 10
0 1 1 2 3 5 8 13 21 34
Process finished with exit code 0
```

4. Prime Number Check



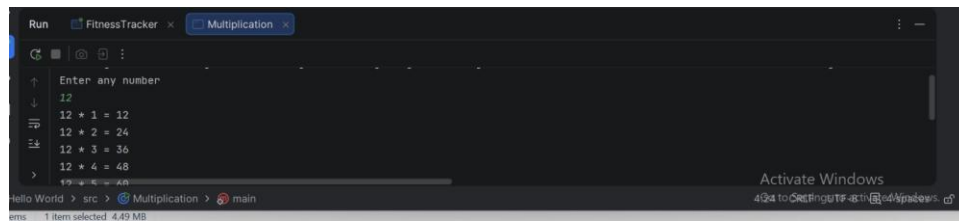
```
Run FitnessTracker PrimeCheck
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57871" -Dfile.encoding=UTF-8
Enter a number: 5
Prime Number
Process finished with exit code 0
```

5. Logical Operators Check



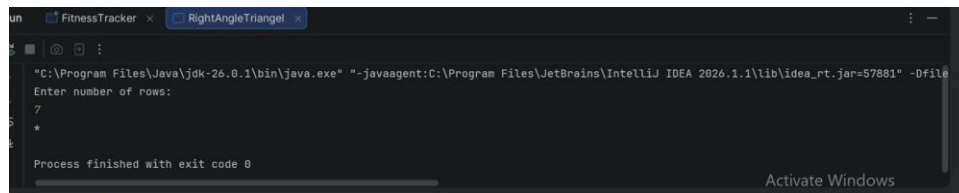
```
Run FitnessTracker LogicalCheck
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57875" -Dfile.encoding=UTF-8
Enter the first number: 15
Enter second number: 23
Enter third number: 25
Third number is not the sum.
Process finished with exit code 0
```

6. Multiplication Table



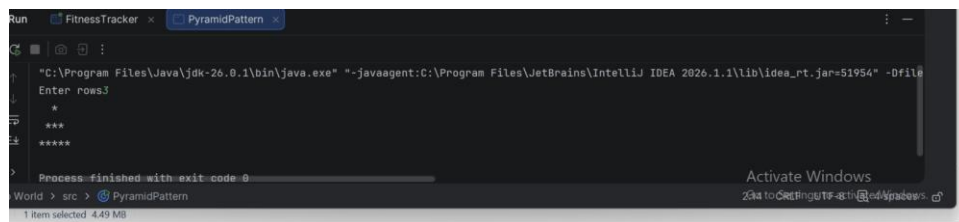
```
Run FitnessTracker x Multiplication x
Enter any number
12
12 * 1 = 12
12 * 2 = 24
12 * 3 = 36
12 * 4 = 48
Process finished with exit code 0
```

7. Right Angle Triangle Pattern



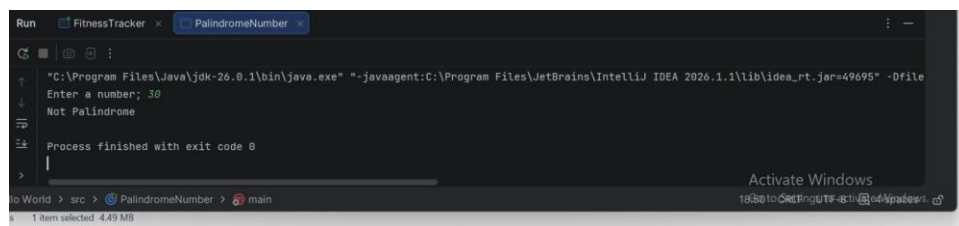
```
Run FitnessTracker x RightAngleTriangle x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=57881" -Dfile
Enter number of rows:
7
*
**
***
****
*****
Process finished with exit code 0
```

8. Pyramid Pattern



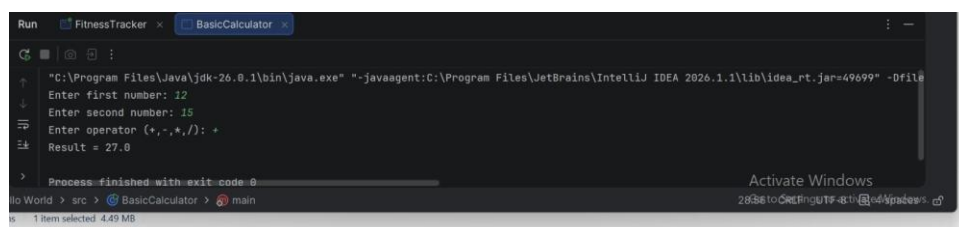
```
Run FitnessTracker x PyramidPattern x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=51954" -Dfile
Enter rows:
3
*
**
***
Process finished with exit code 0
```

9. Palindrome Number



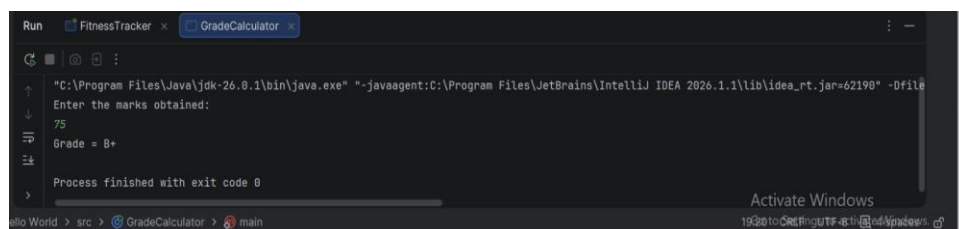
```
Run FitnessTracker x PalindromeNumber x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=49695" -Dfile
Enter a number: 30
Not Palindrome
Process finished with exit code 0
```

10. Basic Calculator



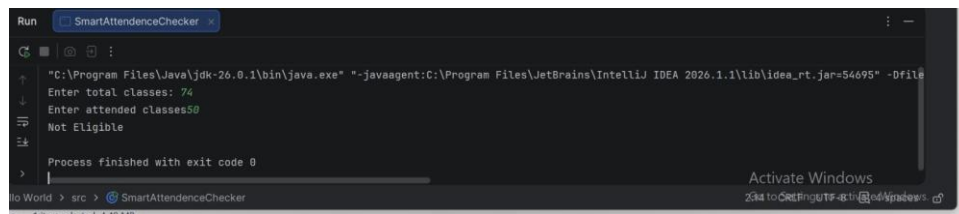
```
Run FitnessTracker x BasicCalculator x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=49699" -Dfile
Enter first number: 12
Enter second number: 15
Enter operator (+, -, *, /): +
Result = 27.0
Process finished with exit code 0
```

11. Grade Calculator



```
Run FitnessTracker x GradeCalculator x
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=62198" -Dfile
Enter the marks obtained:
75
Grade = B+
Process finished with exit code 0
```

12. Smart Attendance Checker



The screenshot shows the Run console of an IntelliJ IDEA project named 'SmartAttendanceChecker'. The console output is as follows:

```
"C:\Program Files\Java\jdk-26.0.1\bin\java.exe" "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA 2026.1.1\lib\idea_rt.jar=54695" -Dfile.encoding=UTF-8
Enter total classes: 74
Enter attended classes: 50
Not Eligible
Process finished with exit code 0
```

At the bottom of the console, there is a message: 'Activate Windows. Go to Settings to activate Windows.' The status bar at the bottom indicates the current file is 'src > SmartAttendanceChecker'.